

Linguistic Echoes: Explainable NLP for Academic Stress Detection

Arjun Pillai

Irvington High School, USA

Project Track: Humanities & Social Science (HSS)

Category: NLP, Mental Health & Education

1 Abstract

Student mental health is one of the most pressing challenges in higher education, yet detection of distress remains largely reactive. This study asks two related questions: can modern NLP systems learn psychologically meaningful stress markers from naturalistic text, and do different academic communities express stress through distinct linguistic patterns?

We trained a TF-IDF Logistic Regression baseline and a fine-tuned DistilBERT transformer on Dreddit, a corpus of 3,553 stress-annotated Reddit posts. Three explainability methods — SHAP, LIME, and LIWC — were applied to surface the linguistic features driving predictions. We then applied these models to 8,000 posts from four academic subreddits (r/college, r/GradSchool, r/PhD, r/PreMed), with Mann-Whitney U tests and Benjamini-Hochberg correction for community comparisons.

DistilBERT achieved 79% accuracy and ROC-AUC 0.874. More importantly, all three explainability methods independently recovered the same markers: elevated first-person singular pronoun use ($\rho = +0.526$ with stress probability, $p < 0.001$) and emotional vocabulary, with social words showing a negative correlation ($\rho = -0.238$). These are markers previously identified in psychology research through entirely different methods, suggesting NLP systems learn theoretically grounded representations rather than arbitrary correlations.

Statistical analysis of the student corpus revealed significant community differences. Undergraduate writers show the highest first-person singular rates (Mann-Whitney U, $p < 0.001$, $d = 0.328$ vs. PreMed), while PhD students show elevated social word usage ($d = -0.126$ vs. college), suggesting they frame distress relationally. All significant findings held after Benjamini-Hochberg false discovery rate correction.

2 Introduction & Research Objectives

2.1 Background

Student mental health has reached crisis levels globally. Over 40% of university students report overwhelming anxiety, and burnout rates among graduate students are substantially higher than in the general population [4]. Most support systems are still reactive — students typically receive help only after significant deterioration, if at all.

Language offers a promising early signal. Research in computational psycholinguistics has demonstrated that writing shifts measurably under stress: stressed individuals use more first-person singular pronouns, more negative affect vocabulary, and show reduced social orientation in their language [2]. These patterns are consistent enough to be measured using lexical

tools like LIWC, but it is less clear whether modern NLP models — trained without any psychological theory — independently recover the same markers. That question is at the center of this study.

A second open question is whether academic communities differ in *how* they express stress. A PhD student and an undergraduate may both experience high stress levels, but describe them very differently. If so, a one-size-fits-all NLP model may systematically miss stress in some communities while over-detecting it in others.

2.2 Research Questions

1. Do explainable NLP methods independently recover psycholinguistic stress markers identified in prior psychology research?
2. What linguistic features most reliably distinguish stressed from non-stressed writing, and how consistent are these across explanation methods?
3. Do different academic communities exhibit statistically distinct linguistic stress profiles?
4. What are the ethical implications of deploying automated stress detection in educational settings?

2.3 Framing

This is not primarily a study about classification accuracy. The classifier is a tool for surfacing linguistic patterns. The main contribution is the combination of multi-method explainability with statistical comparison of how stress is expressed across communities.

Key Contributions

1. **Multi-method convergence:** SHAP, LIME, and LIWC independently recover the same psycholinguistic stress markers, demonstrating that NLP models learn theoretically grounded representations.
2. **Community-level discovery:** First-person singular pronoun use and negative emotion vocabulary differ significantly across four academic communities after Benjamini-Hochberg correction, revealing that stress is not expressed uniformly.
3. **Null finding with implications:** Anxiety vocabulary alone does not differentiate most communities, pointing to self-focus and negative emotion — not explicit anxiety labeling — as the key discriminating signals.
4. **Domain transfer:** A model trained on general stress data transfers meaningfully to academic writing, with the student corpus preserving the same linguistic stress patterns.

3 Methods

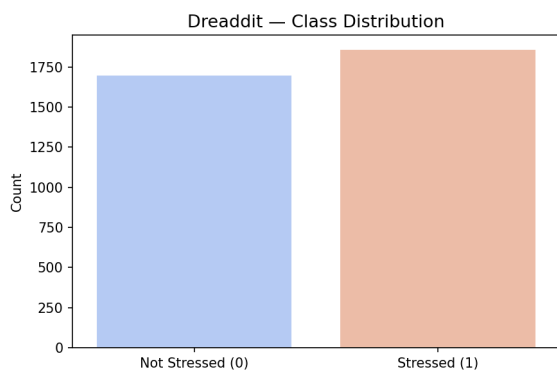
3.1 Overview and Timeline

This project followed a four-stage pipeline over four months. Stage 1 (Month 1): data collection and exploratory analysis — downloading and inspecting the Dreaddit corpus, collecting student Reddit posts via Arctic Shift, and computing LIWC-based EDA. Stage 2 (Month 2): model training — TF-IDF + Logistic Regression locally (VS Code, Windows, NVIDIA RTX 3060) and

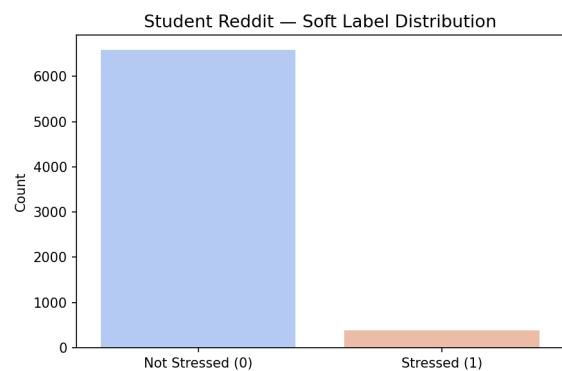
DistilBERT fine-tuning on Kaggle (Tesla T4 GPU). Stage 3 (Month 3): explainability — SHAP and LIME analyses on the trained baseline, followed by domain inference on the student corpus. Stage 4 (Month 4): statistical analysis, error analysis, and paper writing. All code, data, and outputs are version-controlled.

3.2 Datasets

Dreaddit (training corpus). Dreaddit [1] contains 3,553 Reddit post segments from ten subreddits spanning anxiety, PTSD, relationships, domestic violence, and financial hardship. Human annotators labeled each post as stressed (1) or not stressed (0). The dataset is pre-split into 2,838 training and 715 test examples with near-balanced class distribution (52.5% stressed), shown in Figure 1. It includes 93 pre-computed LIWC psycholinguistic dimensions used directly in our analysis.



(a) Dreaddit — near-balanced human-annotated labels



(b) Student corpus — 17:1 soft label imbalance

Figure 1: Class distributions. The 17:1 imbalance from keyword-based soft labeling confirmed our decision to use Dreaddit exclusively for training and student data only for inference and analysis.

Student Reddit corpus (inference and analysis corpus). We collected 2,000 posts each from r/college, r/GradSchool, r/PhD, and r/PreMed via the Arctic Shift public archive, yielding 8,000 posts after filtering for minimum length and removing deleted content. These posts carry no ground-truth stress labels. They were used for domain-specific stress probability inference and direct linguistic feature analysis across communities. Figure 2 shows word count distributions.

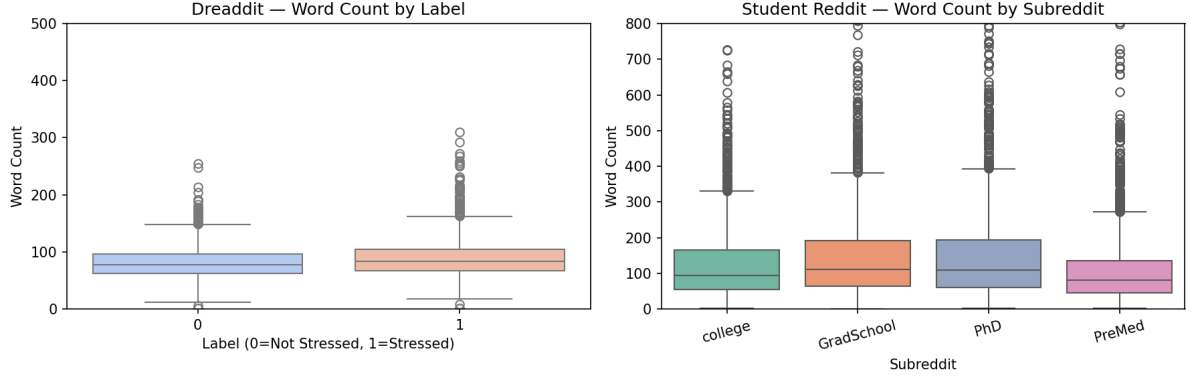


Figure 2: Word count by label (Dreaddit, left) and by subreddit (student corpus, right). PhD and GradSchool posts tend toward longer, more discursive writing.

3.3 Models

TF-IDF + Logistic Regression (baseline). Sparse TF-IDF representation (unigrams and bi-grams, max 10,000 features, min document frequency 3, sublinear TF scaling) with L2-regularized Logistic Regression ($C = 1.0$, balanced class weights).

Fine-tuned DistilBERT. DistilBERT [3] fine-tuned for binary sequence classification over 3 epochs using AdamW ($\text{lr} = 2 \times 10^{-5}$, weight decay 0.01), batch size 32, linear warmup over 10% of training steps. Trained on NVIDIA Tesla T4.

3.4 Explainability Methods

LIWC. Pre-computed LIWC dimensions compared between stressed and non-stressed posts across key psycholinguistic categories.

SHAP. LinearExplainer [5] applied to TF-IDF + LR across the full test set, yielding global importance (mean $|\text{SHAP}|$) and class-conditional attribution profiles.

LIME. [6] Applied to 6 individual examples (3 stressed, 3 not stressed), 500 perturbations each. Weights averaged for a summary view.

3.5 Subreddit Linguistic Analysis and Statistical Methods

For each of the 8,000 student posts we computed four lexicon-based feature rates: first-person singular pronoun rate, anxiety vocabulary rate, negative emotion vocabulary rate, and social words rate. Spearman correlations were computed between all features and the predicted stress probability.

Pairwise community comparisons used Mann-Whitney U tests (non-parametric, appropriate for non-normal rate distributions). To control for multiple comparisons across 15 tests, we applied Benjamini-Hochberg false discovery rate correction at $\alpha = 0.05$. Cohen’s d effect sizes are reported for all comparisons.

4 Results

4.1 Classification Performance

Table 1 reports performance on the Dreaddit test set ($n = 715$).

Table 1: Model performance on the Dreddit held-out test set.

Model	Accuracy	ROC-AUC	F1 (Stressed)	F1 (Not Stressed)
TF-IDF + Logistic Regression	0.720	0.823	0.740	0.706
Fine-tuned DistilBERT	0.790	0.874	0.800	0.770

DistilBERT outperforms the baseline across all metrics. The ROC-AUC of 0.874 reflects strong discriminative capacity across thresholds. Figure 3 shows the confusion matrix and ROC curve.

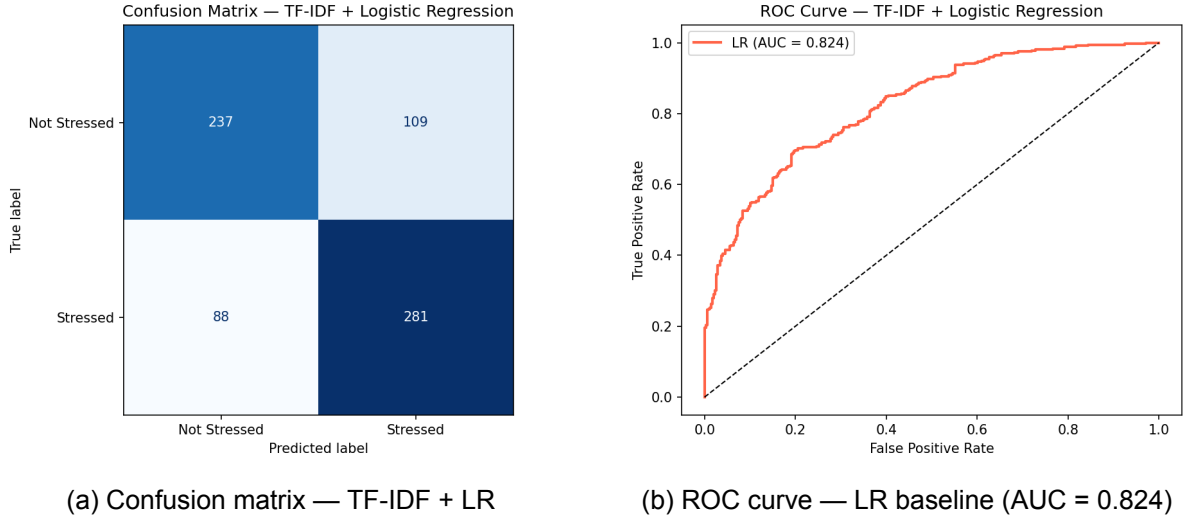


Figure 3: Baseline model evaluation on the Dreddit test set.

4.2 Explainable AI Recovers Psycholinguistic Stress Markers

The central finding is that three independent explainability methods converged on the same linguistic stress markers — and those markers match what psychology research had previously identified through entirely different methods.

LIWC analysis. Figure 4 shows stressed posts have higher scores on anxiety vocabulary, negative emotion, sadness, and anger. The largest difference is in `lex_liwc_i` — first-person singular pronoun use is substantially elevated in stressed posts, consistent with Social Monitoring Theory [2].

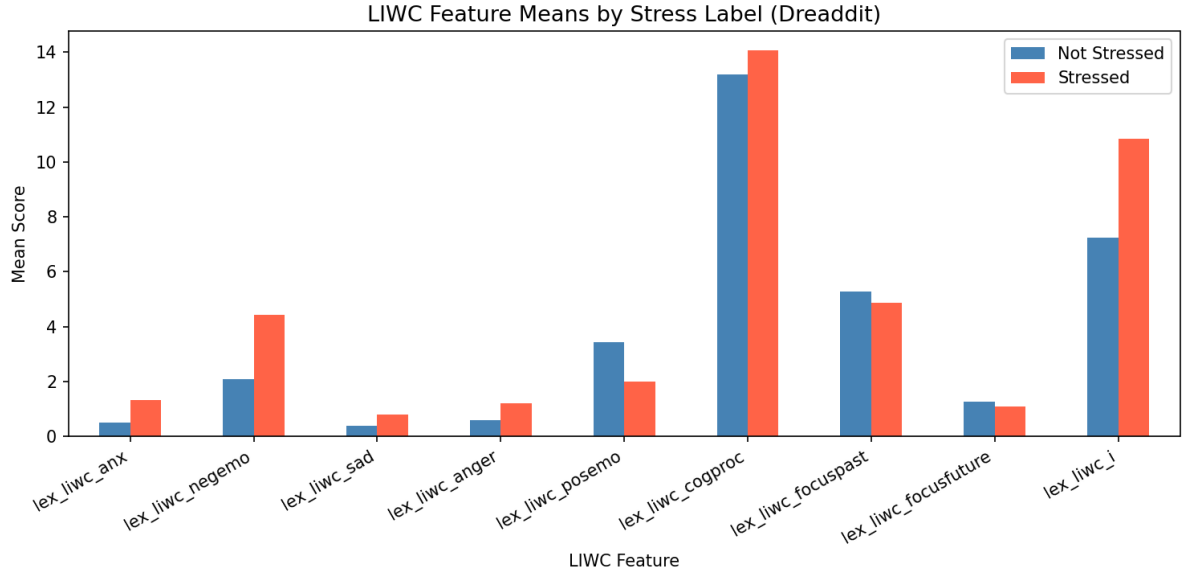


Figure 4: LIWC feature means by stress label. Key differences in `negemo`, `anx`, `posemo`, and `i` align with established psycholinguistic theory.

TF-IDF coefficients. Figure 5 shows the top 20 predictive words. The stressed side is dominated by first-person singular forms (“me”, “my”, “myself”, “am”) and emotional vocabulary (“feel”, “anxiety”, “scared”). The not-stressed side is entirely outward-directed: “you”, “we”, “her”, “thank you”.

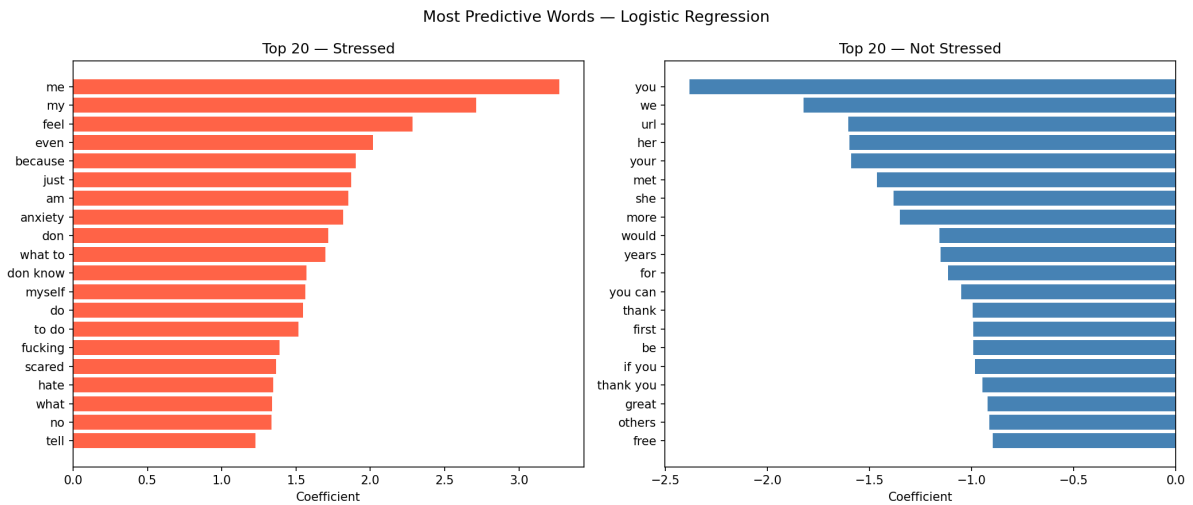
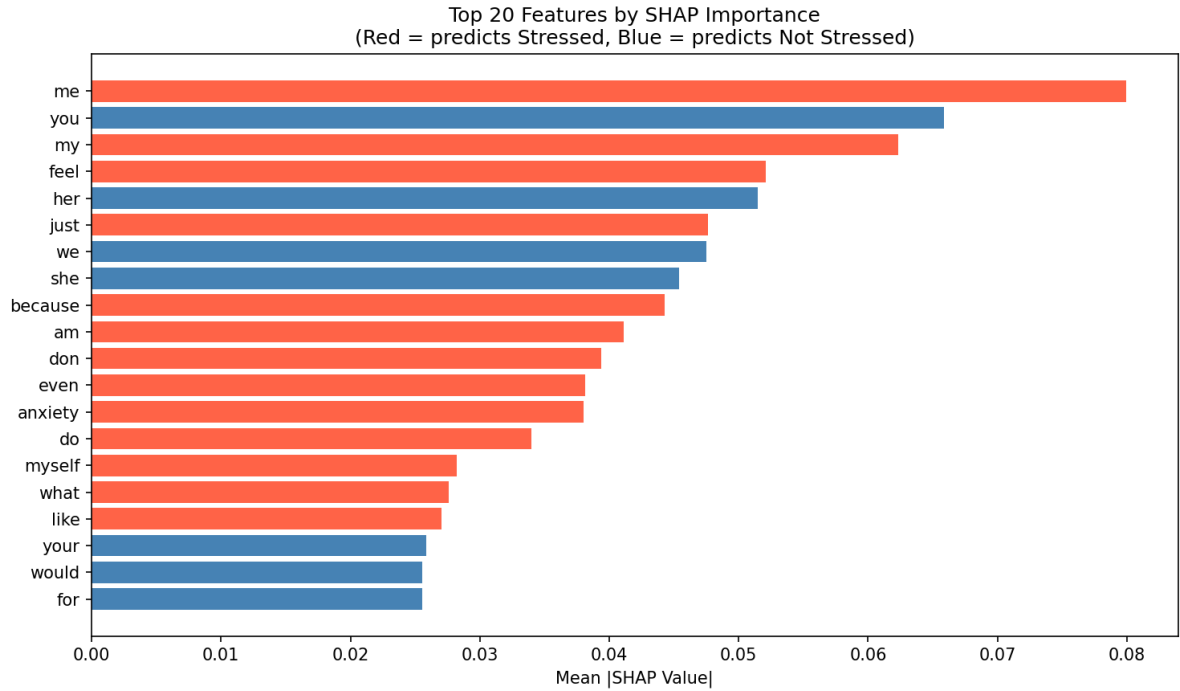
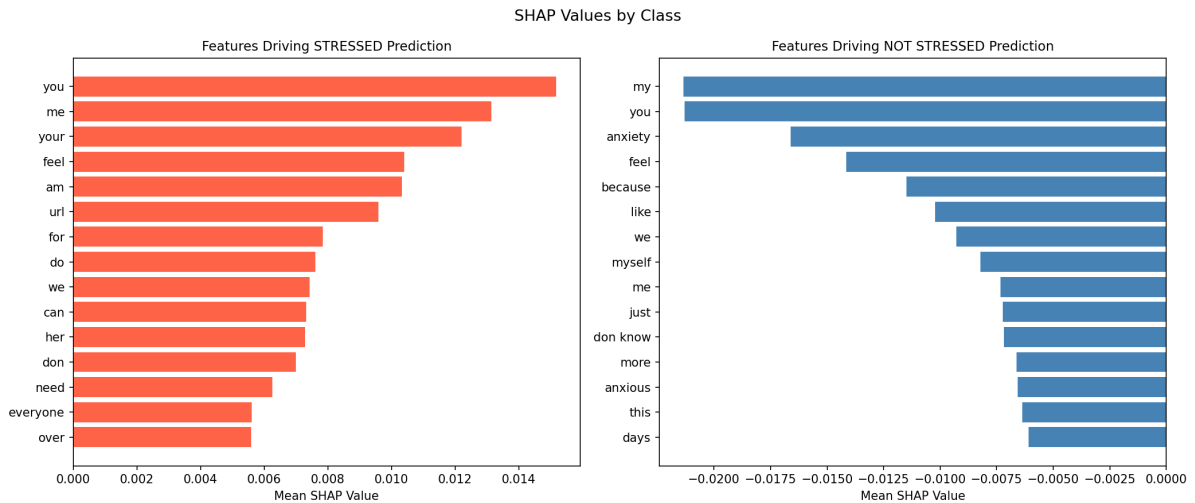


Figure 5: Most predictive words from TF-IDF + LR. Red = stressed; blue = not stressed.

SHAP analysis. Figure 6 shows “me” as the single most important feature globally (mean $|SHAP| = 0.080$), followed by “you” (not-stressed, -0.065), “my”, “feel”, and “anxiety”. The class-conditional breakdown confirms the clean inward/outward split.



(a) Global SHAP importance — top 20 features



(b) Class-conditional SHAP: features driving stressed (left) vs. not-stressed (right)

Figure 6: SHAP feature importance analysis.

LIME analysis. Figure 7 shows averaged LIME weights across six examples. “just”, “me”, “scared”, “myself”, “my”, and “anxiety” receive the highest positive weights; “you”, “would”, “her”, and “we” are the strongest negative drivers. First-person singular markers appear four times independently (“me”, “my”, “myself”, “ME”).

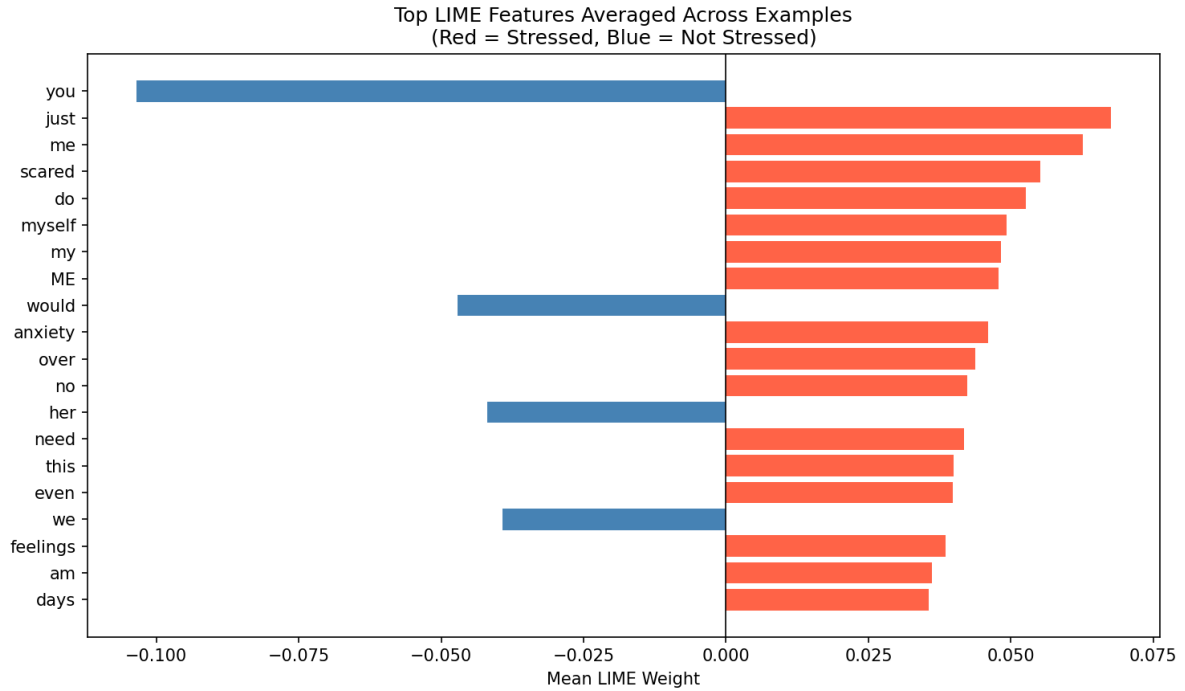


Figure 7: Average LIME feature weights across six examples.

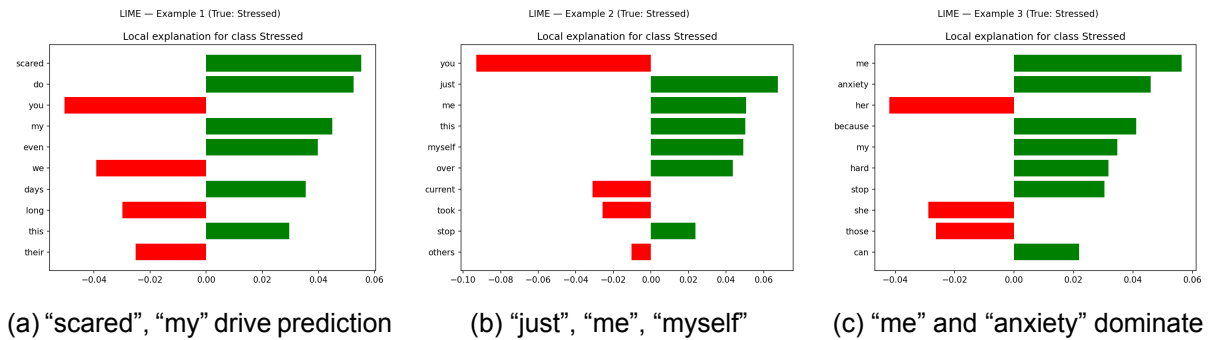


Figure 8: Individual LIME explanations for three stressed examples. Green = pushes toward stressed; red = pushes against.

Table 2 summarises cross-method convergence, with LIWC support column linking each marker to its psycholinguistic counterpart.

Table 2: Three independent methods converge on the same markers, all of which align with psycholinguistic theory.

Feature	TF-IDF	SHAP	LIME	LIWC
me / my / myself	Stressed	Stressed	Stressed	lex_liwc_i ↑
anxiety / scared	Stressed	Stressed	Stressed	lex_liwc_anx ↑
feel / feelings	Stressed	Stressed	Stressed	lex_liwc_negemo ↑
you / we / her	Not Stressed	Not Stressed	Not Stressed	lex_liwc_posemo ↑
would / thank you	Not Stressed	Not Stressed	Not Stressed	Social orientation ↑

Spearman correlation analysis. Figure 9 quantifies relationships between the predicted

stress probability and all linguistic features across the 8,000 student posts. First-person singular rate is the strongest individual predictor ($\rho = +0.526$, $p < 0.001$), followed by negative emotion ($\rho = +0.318$) and anxiety vocabulary ($\rho = +0.282$). Social words show a negative correlation ($\rho = -0.238$), confirming that outward-directed language is associated with lower predicted stress. Notably, social words and negative emotion are essentially uncorrelated with each other ($\rho = 0.007$), supporting the interpretation that these capture distinct dimensions of stress expression.

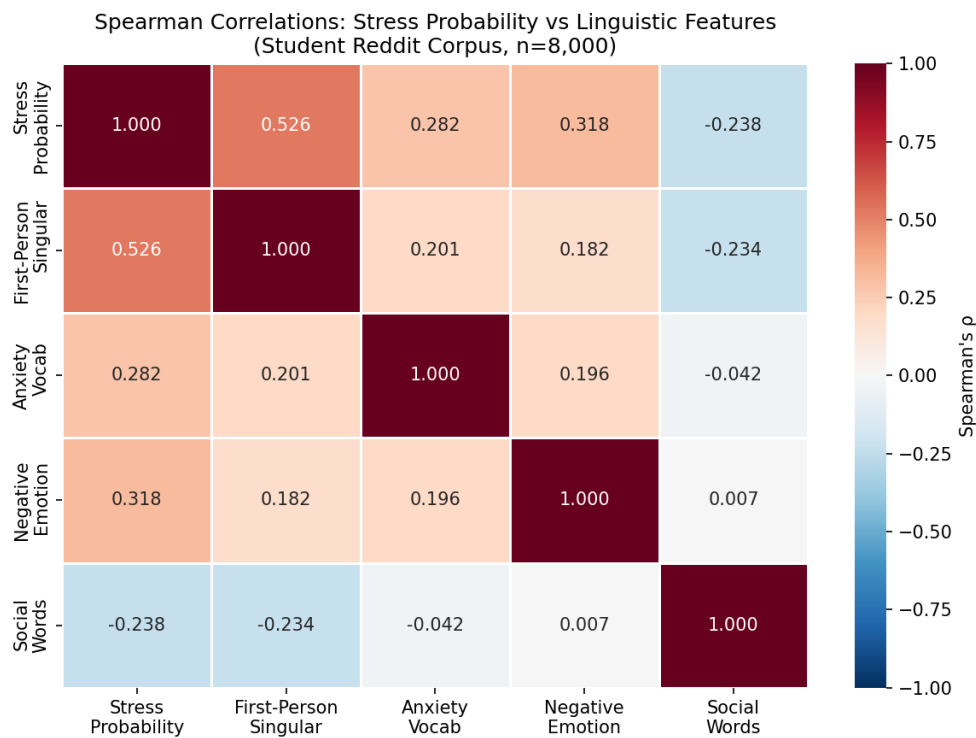


Figure 9: Spearman correlations between predicted stress probability and linguistic features (student corpus, $n = 8,000$). First-person singular rate is by far the strongest predictor. Social words and negative emotion are near-orthogonal ($\rho = 0.007$), suggesting they capture distinct dimensions of stress expression.

Word clouds (Figure 10) show that the academic student corpus preserves the same pattern, with domain-specific terms like “semester”, “classes”, and “research” appearing alongside emotional markers in the stressed cloud.

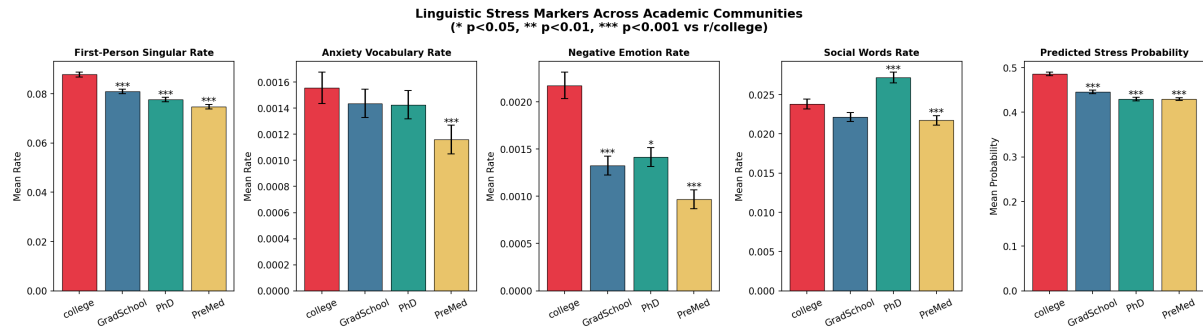


Figure 11: Linguistic stress markers across academic communities. Error bars show standard error. Stars mark Mann-Whitney U tests vs. r/college after Benjamini-Hochberg correction (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

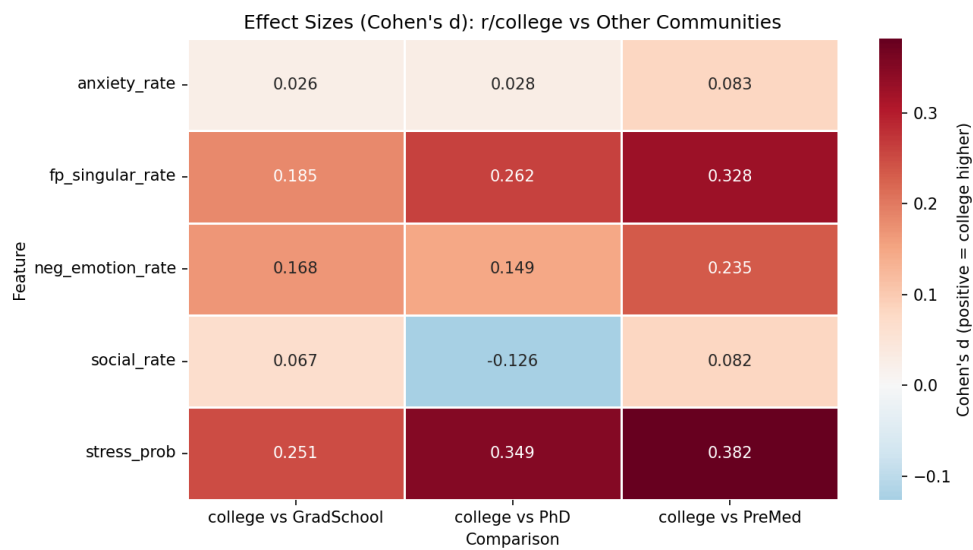


Figure 12: Cohen's d effect sizes: r/college vs. each community. Red = college higher; blue = other community higher. The only blue cell is social_rate vs. PhD.

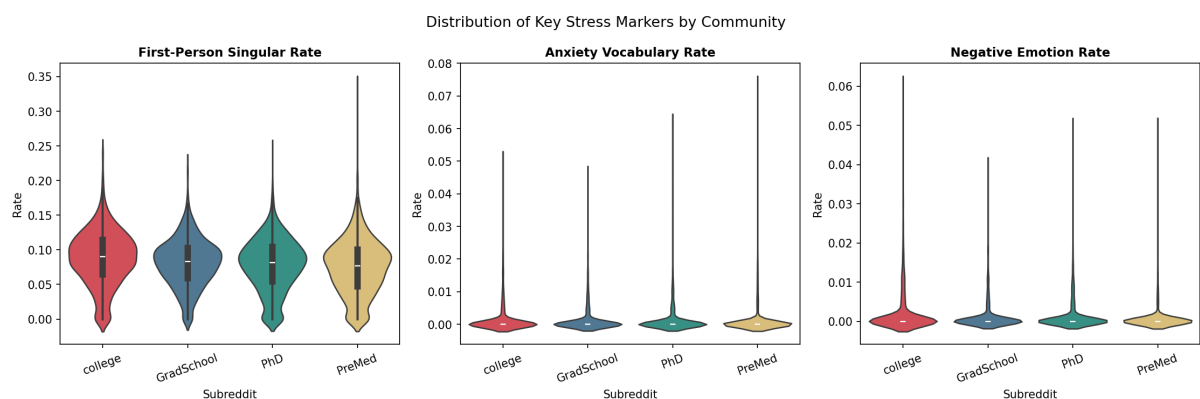


Figure 13: Distributions of key stress markers by community. Within-community variance is high, but medians and tails differ systematically.

Statistical robustness. Fifteen pairwise tests were conducted across five features and three community comparisons. Twelve reached significance at $\alpha = 0.05$. After Benjamini-

Hochberg FDR correction, all twelve remained significant — the correction changed no conclusions. The three non-significant comparisons (anxiety vocabulary vs. GradSchool and PhD; social rate vs. GradSchool) remained non-significant, as expected.

Key findings by feature:

First-person singular rate shows the strongest and most consistent community differences: r/college vs. GradSchool ($p < 0.001$, $d = +0.185$), vs. PhD ($p < 0.001$, $d = +0.262$), vs. PreMed ($p < 0.001$, $d = +0.328$). This is the most robust finding in the analysis.

Negative emotion rate is significantly higher in r/college vs. all communities: vs. GradSchool ($p < 0.001$, $d = +0.168$), vs. PhD ($p = 0.013$, $d = +0.149$), vs. PreMed ($p < 0.001$, $d = +0.235$).

Anxiety vocabulary does *not* significantly differentiate r/college from GradSchool ($p = 0.677$) or PhD ($p = 0.608$) after correction. Only the comparison with PreMed reaches significance ($p < 0.001$, $d = +0.083$, small effect). This is an important null finding: communities are not simply more or less anxious in their word choice — they differ in self-focus and general negative emotion.

Social words: PhD students use significantly more social words than college students ($p < 0.001$, $d = -0.126$, BH-corrected) — the only case where another community scores higher than college on any feature.

Predicted stress probability: All comparisons with college significant ($p < 0.001$, $d = +0.251$ to $+0.382$).

4.4 Error Analysis

Of 715 test predictions, 518 were correct, 109 false positives, and 88 false negatives (Figure 14). The false negative rate of 19.3% is the most critical risk: one in five genuinely stressed posts goes undetected.

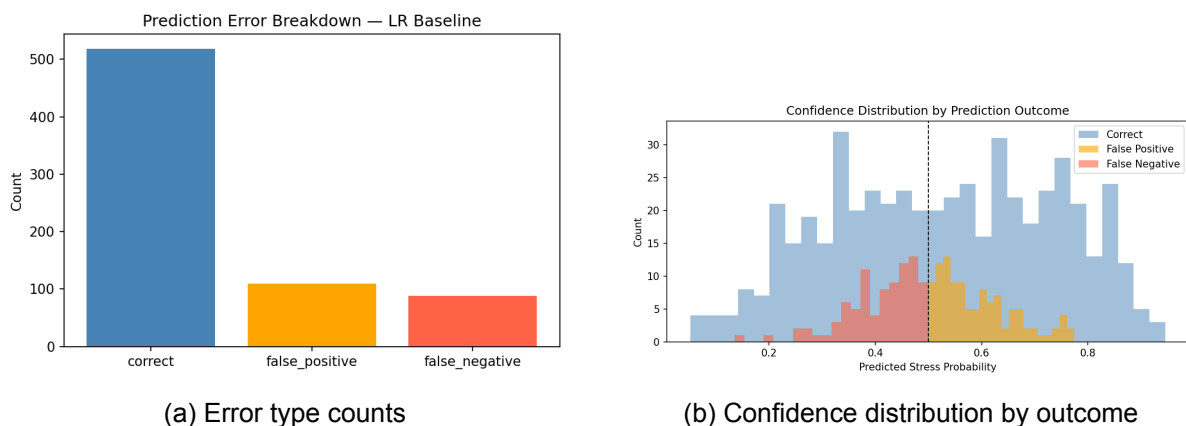


Figure 14: Error analysis on the Dreddit test set.

Qualitative inspection of errors reveals a clear pattern. *False negatives* (stressed posts missed by the model) tend to be written in third-person or past-tense narrative voice, which suppresses the first-person singular signal the model relies on. For example, one false negative from r/relationships (stress prob = 0.136) reads: “*We met about 2.5 years ago... we fell for each other hard... we were both really happy.*” — a genuinely stressful situation described entirely through “we” rather than “I”, making it indistinguishable from non-stressed writing to the classifier.

False positives (non-stressed posts flagged as stressed) typically contain first-person emotional vocabulary in a recovery or retrospective context. For example, one false positive from r/anxiety (stress prob = 0.773) describes *past* social anxiety and recovery from it: “*My flaws*

seemed huge to me... My life was consumed by a spiral of negative thoughts... I went to my doc, admitted my social anxiety, and he prescribed me Lexapro... not two hours after taking it, I felt calmer.” — the model correctly identifies the emotional vocabulary but cannot detect the temporal shift to resolution.

These error patterns suggest two complementary failure modes: the model misses *implicit* stress expressed through third-person narrative, and incorrectly flags *resolved* stress described in first-person retrospective. Both would require richer contextual modeling — potentially temporal or discourse-level features — to address.

Model accuracy varied by subreddit from 63% (r/almosthomeless) to 83% (r/assistance), shown in Figure 15.

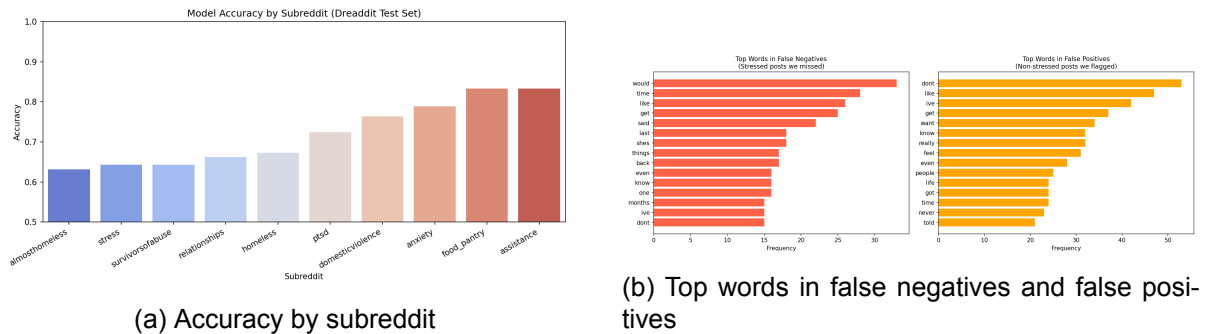


Figure 15: Error analysis by subreddit and word-level breakdown.

5 Discussion

5.1 Explainable AI Recovers Theoretically Grounded Markers

Three independent explainability methods — LIWC comparison, SHAP attribution, and LIME local explanation — all converged on first-person singular pronoun use and negative emotional vocabulary as the primary linguistic stress markers, without psychological theory built into the models. The Spearman correlation of $\rho = 0.526$ between first-person singular rate and predicted stress probability further quantifies this relationship across 8,000 posts.

This convergence matters because it suggests that modern NLP systems learn theoretically meaningful representations rather than arbitrary correlations. The markers recovered here are precisely those predicted by Social Monitoring Theory [2] and measured in clinical psychology through entirely different methods. The social words — negative emotion near-orthogonality ($\rho = 0.007$) further supports the theoretical distinction between inward emotional expression and outward relational framing as separate dimensions of stress-related language.

5.2 Communities Differ in How They Express Stress

The subreddit analysis reveals that different academic communities express stress through linguistically distinct patterns. College students exhibit significantly higher first-person singular rates and negative emotion vocabulary than all other communities, with effect sizes consistent across comparisons ($d = 0.149$ to 0.328). All findings survived Benjamini-Hochberg correction.

PhD students, by contrast, show elevated social word usage compared to undergraduates ($d = -0.126$), suggesting they frame distress in relational terms involving advisors, institutional structures, and peer relationships. Crucially, anxiety vocabulary alone does not differentiate r/college from r/GradSchool or r/PhD. This means communities are not simply more or less anxious in their word choice — they structure distress expression differently.

This has a direct implication for deployment: a stress detection tool trained on general or undergraduate-dominated data may systematically underestimate stress in PhD communities, not because PhD students experience less distress, but because they express it through relational and institutional language that lacks the inward self-focus the model relies on.

5.3 Limitations

Dreaddit is general Reddit data, not academic writing. Domain shift between training and inference populations likely reduces accuracy for student-specific expressions. Effect sizes, while statistically significant, are small to moderate ($d = 0.083$ to 0.382), meaning community membership explains only a fraction of variance. The 19.3% false negative rate is a real deployment risk. Community comparisons use predicted probabilities and lexicon-based features, not ground-truth clinical assessments.

5.4 Responsible AI

Privacy and consent. All data is public Reddit content analyzed at the aggregate level. No individual was identified. Real deployment requires explicit informed consent.

Over-flagging. The TF-IDF baseline produces a 15.2% false positive rate (109 of 715 test predictions), meaning non-stressed students would be incorrectly flagged. Human review before any intervention is non-negotiable.

Expression bias. Accuracy varies 63–83% across subreddits. A tool working better for students who express stress explicitly is inequitable toward communities with relational or suppressed expression.

Scope. This is a research tool for understanding linguistic patterns, not a clinical diagnostic system.

6 Conclusion

This study asked whether explainable NLP can recover psycholinguistic stress markers and reveal how academic communities express distress differently. Both questions have affirmative answers.

Three independent explainability methods converged on first-person singular pronoun use ($\rho = +0.526$ with stress probability) and negative emotional vocabulary as the primary stress markers — the same indicators identified in psychology research through entirely different methods. All significant community differences survived Benjamini-Hochberg false discovery rate correction across 15 tests.

Statistical analysis of 8,000 academic posts showed that undergraduate communities exhibit significantly stronger self-focused and negative-emotion linguistic signals than graduate or pre-medical communities (Mann-Whitney U, $p < 0.001$, d up to 0.382). PhD students show elevated social word usage ($d = -0.126$), suggesting relational framing. Anxiety vocabulary alone does not differentiate most communities, pointing to self-focus and general negative emotion as the key discriminating signals — not explicit anxiety labeling.

Error analysis identified two complementary failure modes: the model misses implicit stress expressed through third-person narrative, and incorrectly flags resolved stress described retrospectively. Future work should collect ground-truth labeled academic writing, test temporal stress modeling across a semester, and investigate whether the relational vs. inward distinction in PhD writing reflects suppression or genuine differences in distress expression.

References

- [1] Turcan, E., & McKeown, K. (2019). Dreaddit: A Reddit Dataset for Stress Analysis in Social Media. *Proceedings of the 9th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, pp. 97–107.
- [2] Pennebaker, J. W., Mehl, M. R., & Niederhoffer, K. G. (2003). Psychological Aspects of Natural Language Use: Our Words, Our Selves. *Annual Review of Psychology*, 54(1), 547–577.
- [3] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. *arXiv preprint arXiv:1910.01108*.
- [4] Evans, T. M., Bira, L., Gastelum, J. B., Weiss, L. T., & Vanderford, N. L. (2018). Evidence for a Mental Health Crisis in Graduate Education. *Nature Biotechnology*, 36(3), 282–284.
- [5] Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*, 30.
- [6] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why Should I Trust You?”: Explaining the Predictions of Any Classifier. *Proceedings of KDD*, pp. 1135–1144.
- [7] Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015). The Development and Psychometric Properties of LIWC2015. *University of Texas at Austin*.